

mHC — Manifold-Constrained Hyper-Connections

Zhenda Xie*, Yixuan Wei*, Huanqi Cao* et al. —  deepseek

Presenter: Amir Sabbaghzariani



Big Picture: Why This Matters

Residual connections are sacred in deep learning

$$\mathbf{x}_{l+1} = \mathbf{x}_l + \mathcal{F}(\mathbf{x}_l, \mathcal{W}_l) \quad \mathbf{x}_L = \mathbf{x}_l + \sum_{i=l}^{L-1} \mathcal{F}(\mathbf{x}_i, \mathcal{W}_i)$$

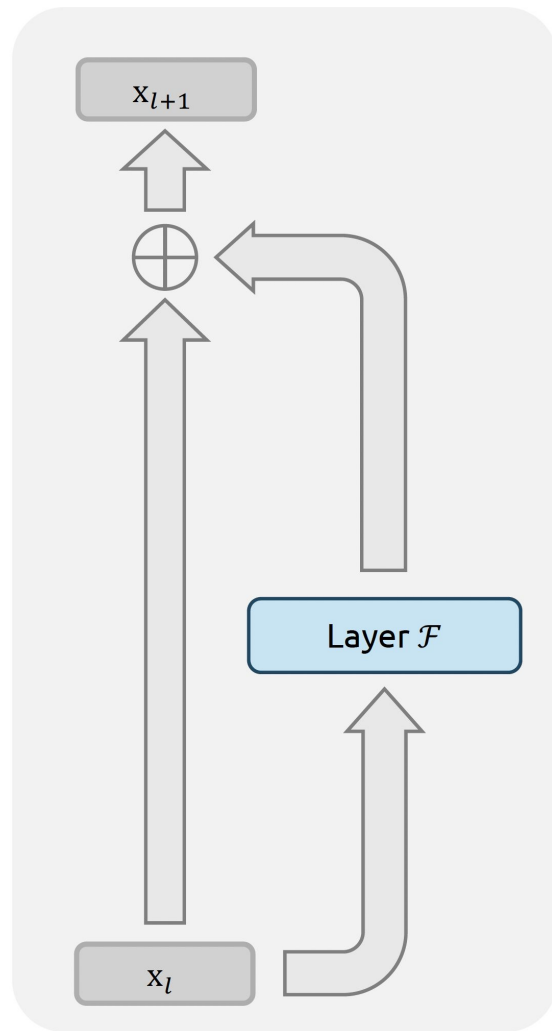
Why they work:

- ❖ Identity mapping enables stable gradient flow
- ❖ Deep signal conservation
- ❖ Scaling to 100+ layers possible

But...

Recent macro-architecture designs expand residual streams.

Question: Can we increase expressivity without breaking identity mapping?



From Residuals to Hyper-Connections (HC)

Hyper-Connections (HC):

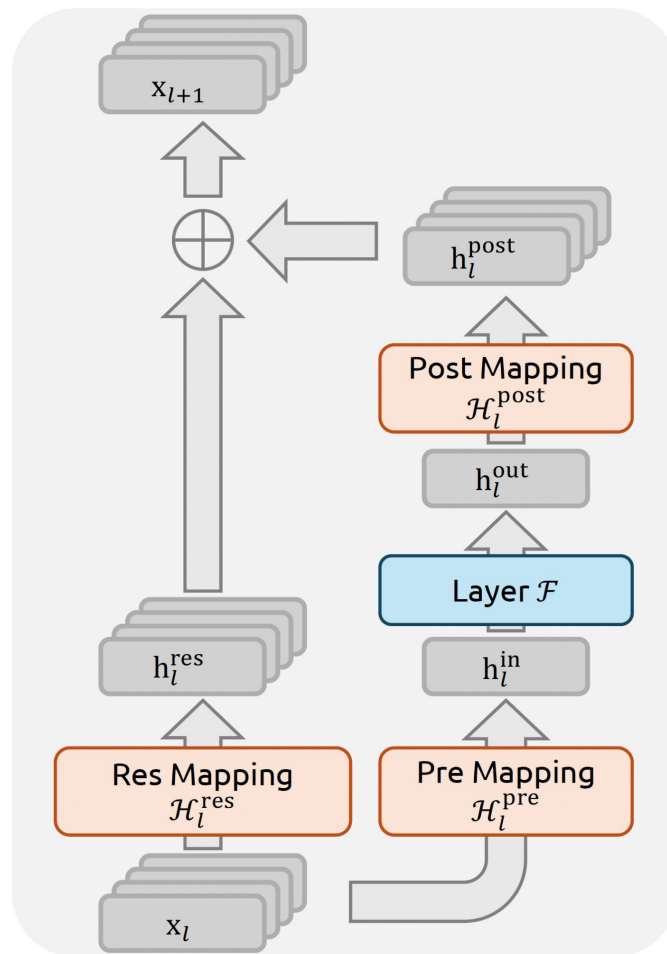
$$\mathbf{x}_{l+1} = \mathcal{H}_l^{\text{res}} \mathbf{x}_l + \mathcal{H}_l^{\text{post} \top} \mathcal{F}(\mathcal{H}_l^{\text{pre}} \mathbf{x}_l, \mathcal{W}_l)$$

HC introduces three mappings:

- ❖ Pre $H_l^{\text{pre}}, H_l^{\text{post}} \in \mathbb{R}^{1 \times n}$
- ❖ Post $H_l^{\text{res}} \in \mathbb{R}^{n \times n}$
- ❖ Residual mixing matrix

Key idea:

- ❖ Expand residual stream from $C \rightarrow nC$
- ❖ Learn how streams mix
- ❖ Increase topology without increasing FLOPs



What HC Buys Us

Benefits:

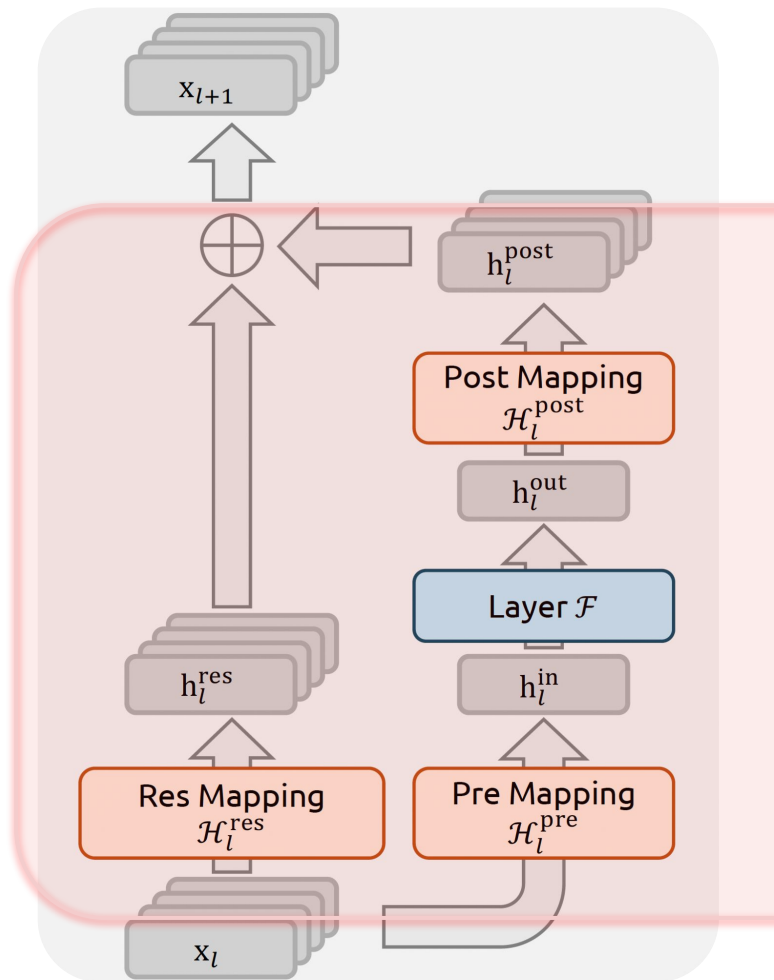
- ❖ Decouples residual width from compute FLOPs
- ❖ Increases routing flexibility
- ❖ Improves performance (shown empirically)



But...

Hyper-Connections removes strict identity mapping.

This is dangerous!



Where Things Go Wrong

HC: $\mathbf{x}_{l+1} = \mathcal{H}_l^{\text{res}} \mathbf{x}_l + \mathcal{H}_l^{\text{post} \top} \mathcal{F}(\mathcal{H}_l^{\text{pre}} \mathbf{x}_l, \mathcal{W}_l)$

Multi-layer HC propagation across layers:

$$\mathbf{x}_L = \left(\prod_{i=1}^{L-1} \mathcal{H}_{L-i}^{\text{res}} \right) \mathbf{x}_l + \sum_{i=l}^{L-1} \left(\prod_{j=1}^{L-1-i} \mathcal{H}_{L-j}^{\text{res}} \right) \mathcal{H}_i^{\text{post} \top} \mathcal{F}(\mathcal{H}_i^{\text{pre}} \mathbf{x}_i, \mathcal{W}_i)$$

The identity path:

In HC:

$$\mathcal{G}_{l \rightarrow L} = \prod_{i=1}^{L-l} H_{L-i}^{\text{res}}$$

$$\|\mathcal{G}_{l \rightarrow L}\| \rightarrow \infty \quad \text{or} \quad 0$$

In ResNet:

$$\mathbf{x}_L = \mathbf{x}_l + \sum_{i=l}^{L-1} \mathcal{F}(\mathbf{x}_i, \mathcal{W}_i)$$

$$\mathcal{G}_{l \rightarrow L} = I$$

$\mathcal{G}_{l \rightarrow L}$ is unconstrained

If unconstrained:

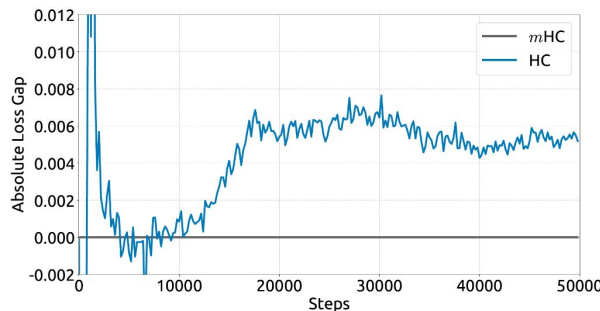
- ❖ Signal amplification
- ❖ Signal attenuation
- ❖ Gradient explosion

Residuals work because identity is preserved.

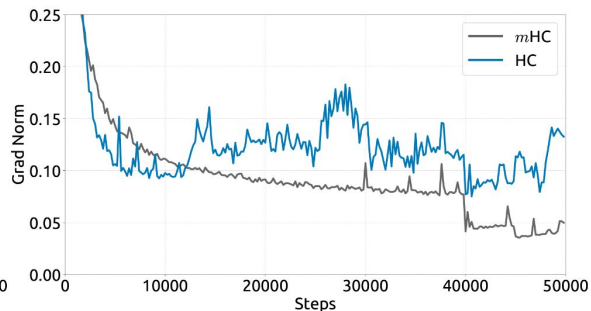
HC destroys that guarantee.

Empirical Instability of HC

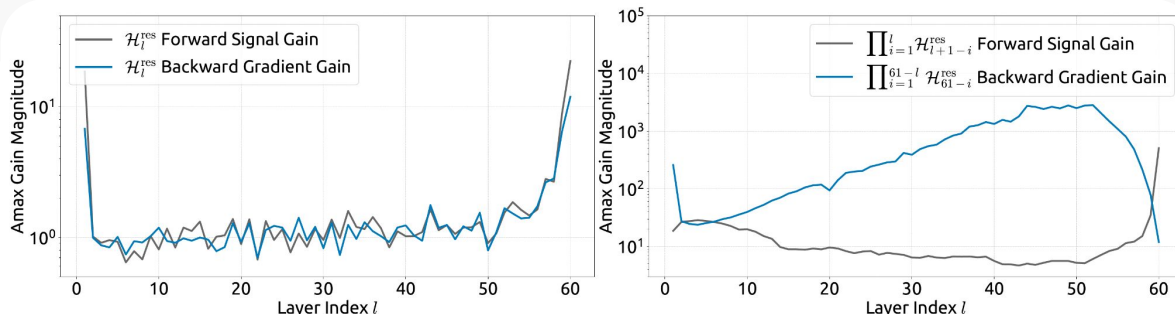
- ❖ Loss spike around 12k steps
- ❖ Gradient norm explosion
- ❖ Composite mapping Amax (maximum absolute signal amplification factor of a matrix) gain peaks **~3000**



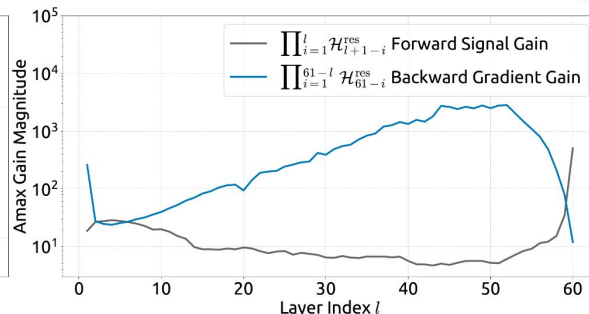
(a) Absolute Training Loss Gap vs. Training Steps



(b) Gradient Norm vs. Training Steps



(a) Single-Layer Mapping



(b) Composite Mapping

What We Need: A Stability Constraint

We want:

- ❖ Information mixing across streams
- ❖ Depth-wise compositional stability
- ❖ Identity mapping generalized

So we search for a manifold:

$$\mathcal{M}_{res} \subset \mathbb{R}^{n \times n}$$

Such that:

- ❖ Closed under multiplication
- ❖ Spectral norm bounded
- ❖ Allows expressivity

This leads to the ***Birkhoff Polytope***

Manifold Definition

Manifold Definition:

$$\mathcal{H}_l^{\text{res}} \in \mathcal{P}_{\mathcal{M}^{\text{res}}}(\mathcal{H}_l^{\text{res}})$$

Defined in:

$$\mathcal{P}_{\mathcal{M}^{\text{res}}}(\mathcal{H}_l^{\text{res}}) := \{ \mathcal{H}_l^{\text{res}} \in \mathbb{R}^{n \times n} \mid \mathcal{H}_l^{\text{res}} \mathbf{1}_n = \mathbf{1}_n, \mathbf{1}_n^\top \mathcal{H}_l^{\text{res}} = \mathbf{1}_n^\top, \mathcal{H}_l^{\text{res}} \geq 0 \}$$

This is the set of doubly stochastic matrices.

The **Birkhoff Polytope** is the convex set of all doubly stochastic matrices (square matrices with nonnegative entries whose rows and columns each sum to one).

Why Doubly Stochastic Works (Spectral Argument)

For any doubly stochastic matrix:

$$\|H_l^{res}\|_2 \leq 1 \xrightarrow{\text{Therefore}} \left\| \prod_{i=1}^{L-l} H_{L-i}^{res} \right\|_2 \leq 1$$

This ensures non-expansive propagation. Thus forward signal gain bounded.

Backward propagation:

$$\nabla x_l = \left(\prod H^{res} \right)^\top \nabla x_L$$

Same bound applies.

This is the mathematical reason Amax drops from **3000 to ~1**.

Compositional Closure

Key geometric property:

If A and B are doubly stochastic,

$A \ B$ is doubly stochastic.

Therefore:

$$\prod_{i=1}^{L-l} H_{L-i}^{res} \in \mathcal{M}_{res}$$

This preserves:

- ❖ Row sums = 1
- ❖ Column sums = 1
- ❖ Non-negativity

Thus identity mapping generalized to convex mixing.

mHC Forward Pass

Flatten residual stream: $\vec{x}_l = \text{vec}(\mathbf{x}_l)$

Dynamic mappings:

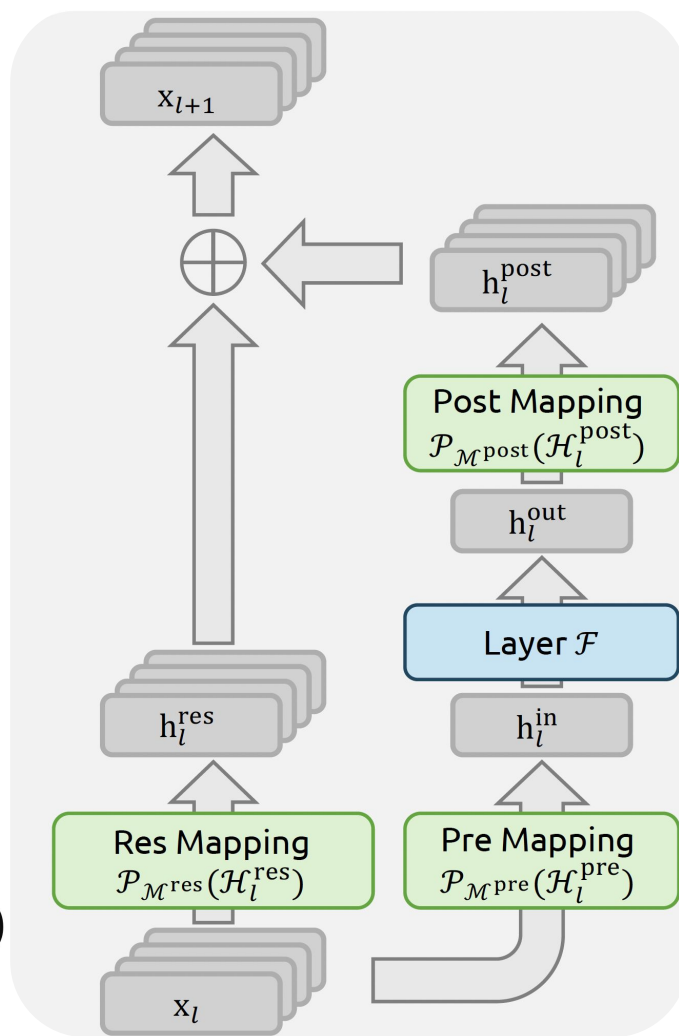
$$\begin{cases} \vec{x}'_l = \text{RMSNorm}(\vec{x}_l) \\ \tilde{\mathcal{H}}_l^{\text{pre}} = \alpha_l^{\text{pre}} \cdot (\vec{x}'_l \varphi_l^{\text{pre}}) + \mathbf{b}_l^{\text{pre}} \\ \tilde{\mathcal{H}}_l^{\text{post}} = \alpha_l^{\text{post}} \cdot (\vec{x}'_l \varphi_l^{\text{post}}) + \mathbf{b}_l^{\text{post}} \\ \tilde{\mathcal{H}}_l^{\text{res}} = \alpha_l^{\text{res}} \cdot \text{mat}(\vec{x}'_l \varphi_l^{\text{res}}) + \mathbf{b}_l^{\text{res}} \end{cases}$$

Then projection:

$$\begin{cases} \mathcal{H}_l^{\text{pre}} = \sigma(\tilde{\mathcal{H}}_l^{\text{pre}}) \\ \mathcal{H}_l^{\text{post}} = 2\sigma(\tilde{\mathcal{H}}_l^{\text{post}}) \\ \mathcal{H}_l^{\text{res}} = \text{Sinkhorn-Knopp}(\tilde{\mathcal{H}}_l^{\text{res}}) \end{cases}$$

Thus final propagation:

$$x_{l+1} = H_l^{\text{res}} x_l + H_l^{\text{post}\top} F(H_l^{\text{pre}} x_l, W_l)$$



Sinkhorn-Knopp Projection

Start with:

$$M^{(0)} = \exp(\tilde{H}_l^{res})$$

Sinkhorn-Knopp Iterative normalization:

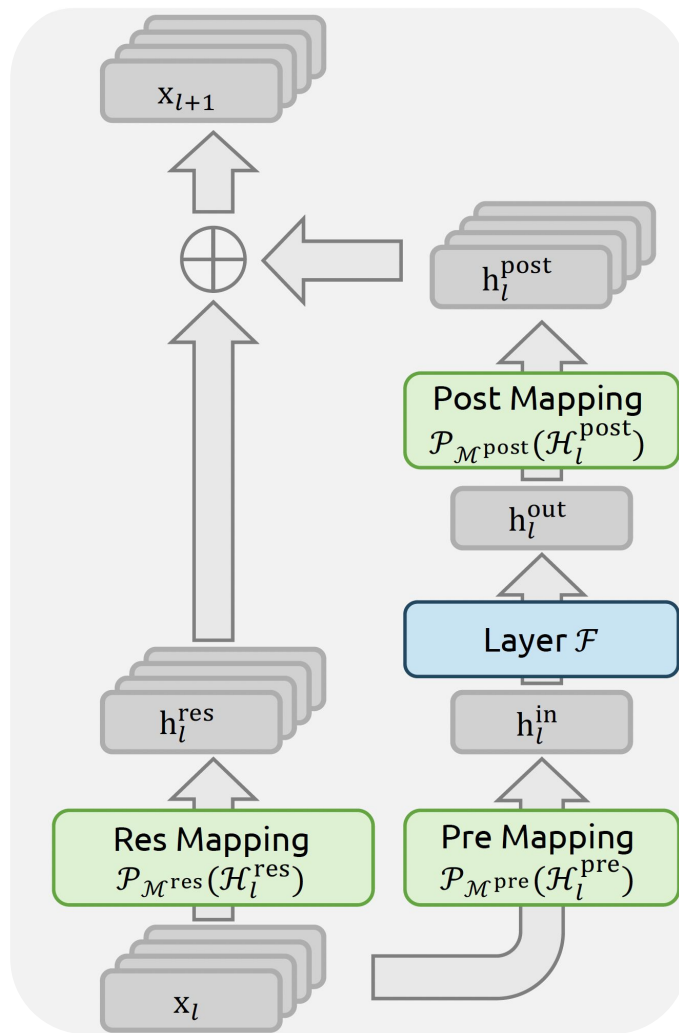
$$M^{(t)} = \underbrace{T_r}_{\text{Row Normalization}}(\underbrace{T_c}_{\text{Column Normalization}}(M^{(t-1)}))$$

$$T_r(M) = \text{diag}(M\mathbf{1})^{-1}M$$

$$T_c(M) = M\text{diag}(\mathbf{1}^\top M)^{-1}$$

After $\mathbf{t}_{\max} = 20$ iterations:

$$H_l^{res} = M^{(t_{max})}$$



Signal Gain Analysis

Ideal condition:

$$H_l^{res} \mathbf{1} = \mathbf{1}$$

So forward gain:

$$\max_i \sum_j H_{ij}^{res} = 1$$

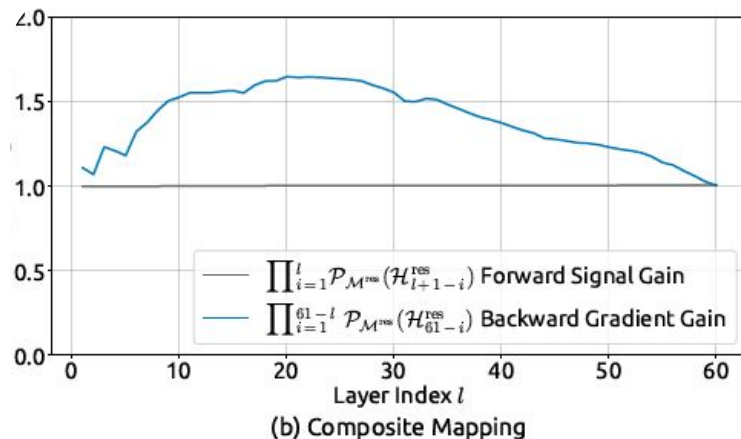
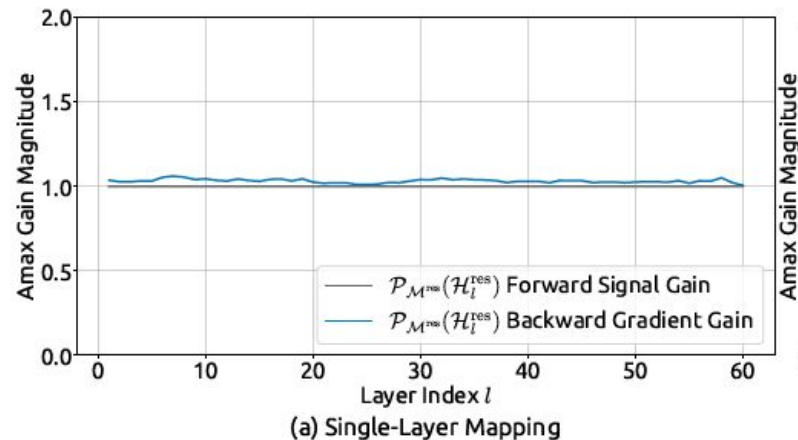
Backward gain:

$$\max_j \sum_i H_{ij}^{res} = 1$$

Empirically: Composite gain ≤ 1.6

HC:

❖ Composite gain ≈ 3000



Comparison

HC:

- ❖ High expressivity
- ❖ No spectral constraint

Residual:

- ❖ Full stability
- ❖ No mixing

mHC:

- ❖ Convex mixing
- ❖ spectral bound

Thus:

Manifold constraint acts as geometric regularizer.

Conceptual Takeaway:

mHC reframes residual architecture as a depth-wise Markov mixing process constrained on a manifold.